



AWS IAM Hands-On Lab Guide

A step-by-step walkthrough for inspecting IAM configurations, validating policy-driven access controls, and executing user group assignments in a live AWS sandbox environment.

IT STUDENTS & EARLY-CAREER CLOUD ENGINEERS

40-MINUTE LAB

Lab Overview & Learning Objectives

This hands-on exercise places you inside a real AWS environment where IAM configurations are already pre-built. Your job is to audit them, assign users to groups, and test what those permissions actually allow — or deny.



Explore IAM Users & Groups

Review pre-created users and functional group structures inside the AWS console.



Inspect JSON Policies

Read and interpret both Managed and Inline policy documents attached to groups.



Assign Users to Groups

Follow a corporate business scenario to map each user to their appropriate access group.



Validate Permission Boundaries

Log in as each user and confirm that access restrictions behave exactly as expected.

Prerequisites & Setup

What You Need

- **Environment:** AWS Academy Guided Lab (Course Pre-Packaged)
- **Time Limit:** 40 minutes active runtime
- **Browser:** Chrome, Firefox, or Edge – pop-up blockers **disabled**

What's Pre-Built for You

The lab backend automatically provisions the following resources before you begin:

user-1

Placeholder user

user-2

Placeholder user

user-3

Placeholder user

EC2-Admin

EC2-Support

S3-Support

PHASE 1

Accessing the Lab Environment

1 Start the Lab

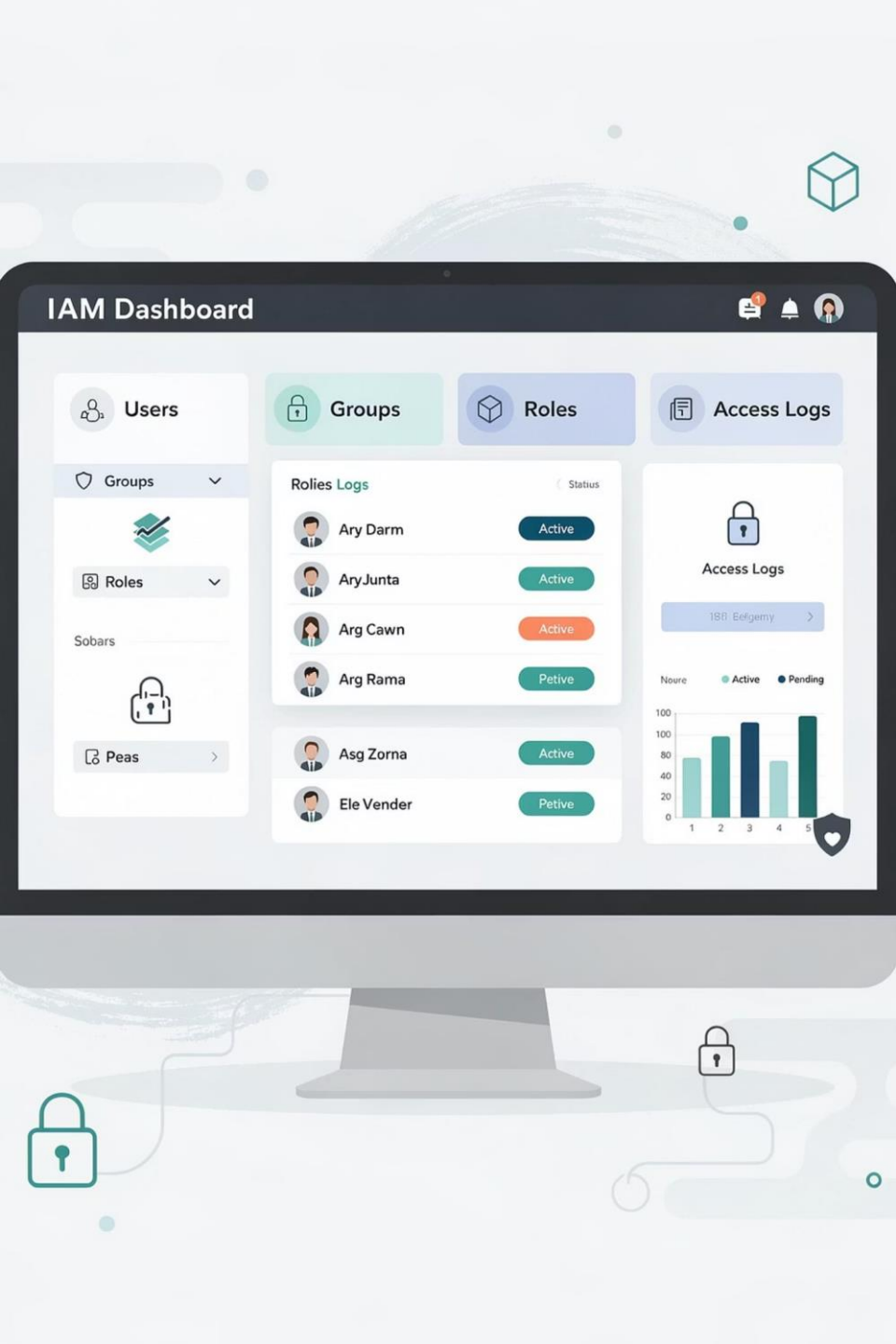
Click the **Start Lab** button at the top of your lab portal. A countdown timer will appear in the header, showing your remaining session time. You can click **Start Lab** again before it hits 0:00 to safely extend your session.

2 Wait for Green Status

Watch the status indicator icon to the right of the **AWS** link in the upper-left corner. Wait until it transitions from **amber** to a solid **green** — this confirms your sandbox is fully provisioned and ready.

3 Open the AWS Console

Click the **AWS** text link. A new browser tab opens and logs you directly into the AWS Management Console. Arrange this tab side-by-side with your lab instructions so you can follow along without constant window switching.



PHASE 2

Auditing Users, Groups & JSON Policies

Before making any changes, you'll inspect the pre-built IAM environment — understanding what exists and why — to build a solid foundation for the assignments ahead.

Navigating to IAM & Reviewing Users

01

Open IAM

Use the top search bar (to the right of the **Services** menu), type **IAM**, and select it from the filtered results to open the Identity and Access Management dashboard.

02

Select Users

In the left-hand sidebar, click **Users**. You'll see three pre-provisioned accounts: **user-1**, **user-2**, and **user-3**. Click on **user-1** to open their profile.

03

Review user-1's Tabs

On the **Permissions** tab, notice **no policies are assigned**. On the **Groups** tab, confirm **no group membership** exists yet. On **Security credentials**, note that a **Console password** is configured — this user can log in to the AWS web console.

 All three users start with zero permissions. Permissions will be inherited once they are added to their respective groups in Phase 3.

Inspecting the EC2-Support Group Policy

Return to the left sidebar, click **User groups**, and open the **EC2-Support** group. On the **Permissions** tab, you'll find an AWS Managed Policy: **AmazonEC2ReadOnlyAccess**. Click the **+** icon to expand the JSON.

```
{
  "Effect": "Allow",
  "Action": [
    "ec2:Describe*",
    "ec2:Get*",
    "cloudwatch:ListMetrics",
    "elasticloadbalancing:Describe*",
    "autoscaling:Describe*"
  ],
  "Resource": "*"
}
```

This policy grants **read-only** access: users can list and describe EC2 resources and view CloudWatch metrics, but they **cannot modify or delete** anything. The wildcard `Describe*` pattern covers all describe-type API calls. Click the **–** icon to collapse the view when done.

Inspecting the S3-Support Group Policy

Return to **User groups** in the left sidebar and open the **S3-Support** group. On its **Permissions** tab, expand the **AmazonS3ReadOnlyAccess** managed policy using the + toggle icon.

This policy restricts members strictly to **Get** and **List** actions on Amazon S3 objects and buckets. No upload, delete, or modify operations are permitted. This is a classic example of the **Principle of Least Privilege** – granting only the minimum access needed for a role.

Key Policy Actions Allowed



s3:GetObject

Download/view object contents



s3:ListBucket

View bucket contents & object names



s3:GetBucketPolicy

Read existing bucket policies

Inspecting the EC2-Admin Group Policy

Return to **User groups** and open the **EC2-Admin** group. In its **Permissions** tab, notice something different: this group uses an **Inline Policy** rather than an AWS Managed Policy. Inline policies are embedded directly into a single entity and are not reusable across other groups.

```
{
  "Effect": "Allow",
  "Action": [
    "ec2:Describe*",
    "ec2:StartInstances",
    "ec2:StopInstances"
  ],
  "Resource": "*"
}
```

Describe*

View and list all EC2 resources

StartInstances

Power on EC2 instances

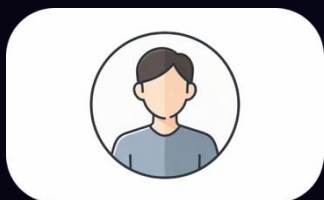
StopInstances

Gracefully shut down EC2 instances

PHASE 3

Enforcing the Corporate Business Scenario

You will now assign each user to their designated group, granting them exactly the access their role requires — no more, no less. This mirrors how real enterprise cloud teams structure access control.



user-1 → S3-Support

Read-only access restricted exclusively to Amazon S3 data stores.



user-2 → EC2-Support

Read-only monitoring restricted to Amazon EC2 compute systems.



user-3 → EC2-Admin

Full view, start, and stop administrative actions on EC2 instances.

Step 3.1 & 3.2: Adding user-1 and user-2 to Groups

user-1 → S3-Support

1. In the left sidebar, click **User groups** and open **S3-Support**.
2. Click the **Users** tab, then click **Add users**.
3. Check the box next to **user-1**.
4. Click **Add users** at the bottom to confirm. Verify **user-1** now appears in the membership table.

user-2 → EC2-Support

1. Return to **User groups** in the left sidebar. Open **EC2-Support**.
2. Click the **Users** tab, then click **Add users**.
3. Check the box next to **user-2**.
4. Click **Add users** to confirm. Verify **user-2** appears in the group membership log.

Step 3.3: Adding user-3 to EC2-Admin

01

Open EC2-Admin

Click **User groups** in the left sidebar. Select the **EC2-Admin** group name link.

02

Add user-3

Open the **Users** tab and click **Add users**. Check the box next to **user-3**. Click the **Add users** confirmation button.

03

Verify All Assignments

Return to the main **User groups** view via the left sidebar. Check the **Users** column for all three groups. Every group must display a count of **1** — confirming all assignments are complete.



Once all three groups show a count of 1, your assignment phase is complete. Each user now inherits their group's permissions automatically.

PHASE 4

Testing & Validating Permission Isolation

To properly verify the Principle of Least Privilege, you must test each user in a fresh private/incognito browser window. This prevents cached credentials from one session polluting another.



Step 4.1: Retrieve the Sign-In URL

1 Open the IAM Dashboard

In the left sidebar of the IAM console, click **Dashboard** at the very top. On the right side of the page, locate the section labeled "**Sign-in URL for IAM users in this account.**"

2 Copy the URL

The URL will look like:

```
https://123456789012.signin.  
aws.amazon.com/console.
```

Highlight and copy it to your clipboard or a temporary text editor scratchpad. You'll use this same URL for all three user logins.

3 Open a Private Window

Launch a new **Private / Incognito** browser window (**Ctrl+Shift+N** on Windows, **Cmd+Shift+N** on Mac for Chrome). Paste the copied URL into the address bar and press **Enter**.

Step 4.1 (cont.): Validate user-1 Access

Log in with **IAM user name: user-1** and **Password: Lab-Password1**. Then run the following validation checks:

S3 Access — ALLOWED

Search for **S3** in the top bar and open the S3 console. Click the pre-existing bucket name to browse its contents. You should successfully see folder layers — this confirms the **S3-Support** group policy is active.

EC2 Access — BLOCKED

Search for **EC2**, open the console, and click **Instances** in the left sidebar. The page must display: "**You are not authorized to perform this operation.**" This confirms cross-service isolation is working correctly.

Once validated, click the **user-1** dropdown in the top-right header and select **Sign Out**.

Step 4.2: Validate user-2 Access (EC2-Support)

Paste the IAM sign-in URL back into the private browser. Log in with **IAM user name: user-2** and **Password: Lab-Password2**. Run these validation checks:

EC2 View — ALLOWED

Search for **EC2** and click **Instances**.
Verify the **LabHost** instance is visible in the grid. If no instance appears, check the region dropdown (top-right) and confirm it is set to **N. Virginia**.

Stop Instance — BLOCKED

Select **LabHost**, click **Instance state** → **Stop instance**, and confirm. The action must fail with: **"You are not authorized to perform this operation."** Read-only policy working as designed.

S3 Access — BLOCKED

Search for **S3**. The dashboard must display: **"You don't have permissions to list buckets."** This confirms user-2 is fully restricted to EC2 read functions only.

Sign out via the **user-2** dropdown at the top-right header when done.

Step 4.3: Execute Admin Actions as user-3

Paste the IAM sign-in URL into the private browser. Log in with **IAM user name: user-3** and **Password: Lab-Password3**.

01

Open EC2 Instances

Search for **EC2**, open the console, and click **Instances** in the left sidebar. Locate the **LabHost** instance row.

02

Stop the Instance

Check the box next to **LabHost**. Click **Instance state** → **Stop instance**. Click the **Stop** button in the confirmation prompt.

03

Confirm the Result

Watch the status column in the instance grid. The server must transition to a blinking amber **Stopping** state — confirming the EC2-Admin inline policy is fully active and functioning. Close the private browser window when done.

PHASE 5

Submitting & Cleaning Up

Submit Your Score

1. Return to your original lab instructions browser tab.
2. Click the **Submit** button in the top header to record your lab score.
3. If points don't appear immediately, wait **2 minutes** and click Submit again — backend automated checks need a buffer window of up to 5 minutes after resource changes.

End the Lab

1. Once your grade is confirmed, click the **End Lab** button at the top of the interface.
2. Click **Yes** in the confirmation prompt to cleanly terminate your sandbox environment.



Do not close your browser tab before clicking End Lab. Skipping this step may leave your session open and consume your remaining lab time.

Troubleshooting Common Issues

Symptom / Error	Root Cause	Fix
"Failed to stop instance: Not authorized"	Logged in as user-1 or user-2, neither has stop permissions.	This is expected. Sign out and log in as user-3 .
"You don't have permissions to list buckets"	Logged in as user-2 or user-3 – S3 is not in their policy scope.	This is expected. Use user-1 to access S3.
Blank instance grid / LabHost not visible	Console region has defaulted away from the lab deployment zone.	Click the region dropdown (top-right) and set it to N. Virginia .
Sign-in URL loads blank or errors	Browser extension or proxy is blocking the IAM redirect path.	Clear cache, use a fresh Incognito window, or try a different browser.

Lab Summary & Key Takeaways

You've successfully completed all five phases of this AWS IAM lab. Here's what you demonstrated in a real cloud environment:

IAM Structure

Users inherit permissions through group membership — not direct policy attachment. Groups are the scalable way to manage access at enterprise scale.

Policy Types

AWS Managed Policies are reusable and maintained by AWS. Inline Policies are custom and bound to a single entity — used when a unique, non-shareable permission set is needed.

Least Privilege

Each user could only perform actions explicitly granted. Attempts to exceed those permissions were blocked immediately — exactly as intended by sound security design.

Testing Isolation

Using private/incognito windows for each user ensures clean credential testing — a critical real-world practice when validating IAM policy behavior.

